

	Case Name: Railway Station (1)	Sector	Construction (civil)	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
			One of nine std. scenarios	
Submitted by	Matilde Casado and Margarida Antunes		User defined distributions	
Date	2015		Automatic tracking	
File Name	C2015-02 Railway Station (1)		Tracking based on user input	

1. Project description

Project authenticity

The project involves the demolition of old structures, extensive groundwork, deep foundations, structural steel installations, platform construction, roofing, drainage, and integration of electrical and special technologies, ensuring a modernized and efficient transportation hub.

The project consists of activity data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	216
Planned Duration (PD)	417 days*
Budget At Completion (BAC)	1 121 316.9€
Renewable Resources	-
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	8%
Activity Distribution (AD)	66%
Length of Arcs (LA)	1%
Topological Float (TF)	80%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	3.7	7.8	1.0
CRI-rho	32.3	21.5	1.0
CRI-tau	59.6	46.8	1.0

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	5.5	11.1	1.0
SI	17.5	26.7	N/A
SSI	3.0	7.5	N/A
CRI-r	8.6	6.6	-1.0
CRI-rho	8.5	6.8	-1.0
CRI-tau	9.7	6.8	-1.0

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	19.9	19.9	1	1.5	0.2
PV - SPI	28.8	28.8	CPI	3.1	0.9
PV - SCI	29.4	29.2	SPI	10.9	10.5
ED - 1	18.0	18.0	SPI(t)	10.2	9.7
ED - SPI	27.3	27.3	SCI	11.2	10.7
ED - SCI	27.4	27.4	SCI(t)	10.5	9.9
ES - 1	15.7	15.7	0.8 CPI + 0.2 SPI	4.6	4.0
ES - SPI(t)	24.4	24.3	0.8 CPI + 0.2 SPI(t)	4.5	3.8
ES - SCI(t)	24.5	24.4			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 9 tracking periods with a length of approximately 2 to 3 months. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-70399.6	-332996.1	-961.0	0.8	0.6	0.6	0.9
std dev	92 224.7	247 892.0	546.9	0.1	0.2	0.2	0.1
final	153 329.0	0.0	-672.0	1.2	1.0	0.8	1.0

2.3.3.2. Time forecasting

PD	417 days	Real Duration	501 days	Late	20.1%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	432.2	2.1	1.5	1.5
PV - SPI	432.8	2.1	1.5	1.5
PV - SCI	433.4	2.1	1.5	1.5
ED - 1	434.2	1.9	1.5	1.5
ED - SPI	435.0	1.9	1.5	1.5
ED - SCI	435.8	1.9	1.4	1.4
ES - 1	436.6	2.1	1.4	1.4
ES - SPI(t)	437.2	2.1	1.4	1.4
ES - SCI(t)	437.8	2.1	1.4	1.4

2.3.3.3. Cost forecasting

BAC	1 121 316.9€	Real Cost	967 988.3€	On Budget	-13.7%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	1191716.8	92 224.7	2.6	-2.6
CPI	1420846.1	215 687.6	5.2	-5.2
SPI	1807433.6	567 838.2	9.6	-9.6
SPI(t)	1849335.1	487 065.8	10.1	-10.1
SCI	2134364.0	911 154.8	13.4	-13.4
SCI(t)	2158210.8	833 569.0	13.7	-13.7
0.8 CPI + 0.2 SPI	1476670.5	249 317.1	5.8	-5.8
0.8 CPI + 0.2 SPI(t)	1477349.6	250 045.4	5.8	-5.8